

<https://introml.mit.edu/>

6.390 Intro to Machine Learning

Lecture 4: Linear Classification

Shen Shen

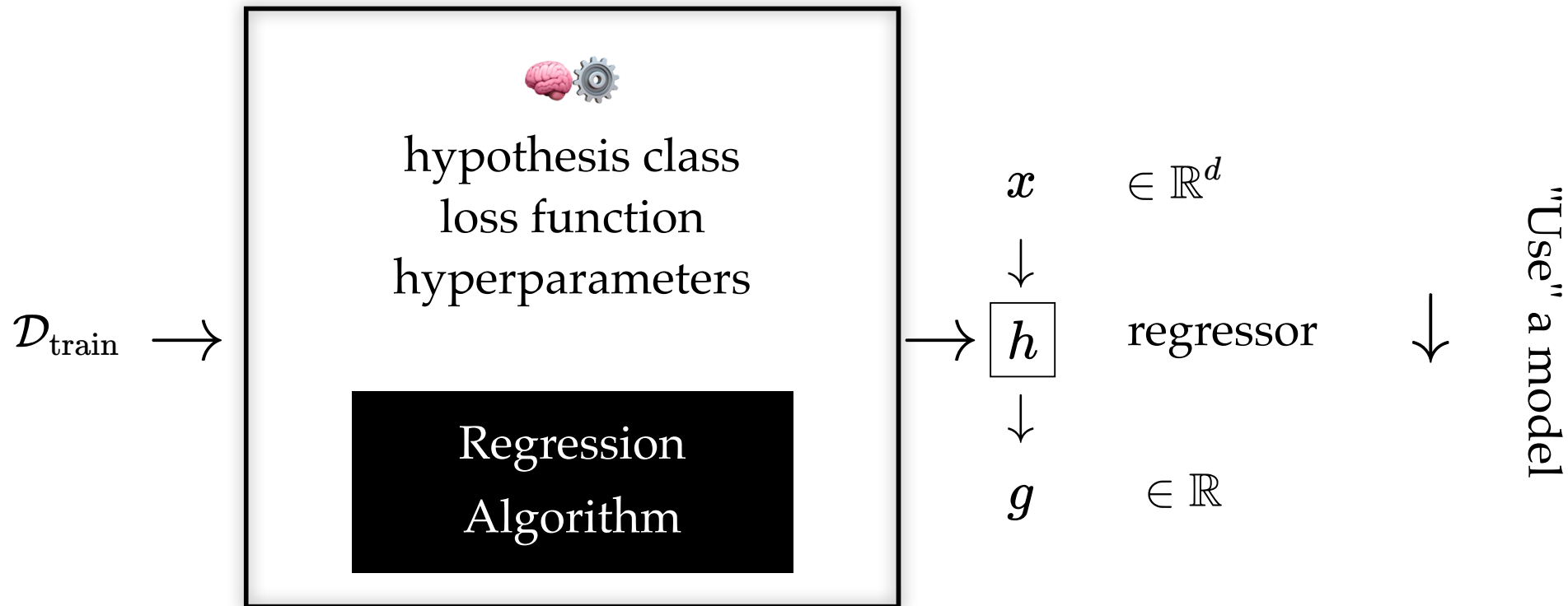
Feb 23, 2026

Async (Snow Day)

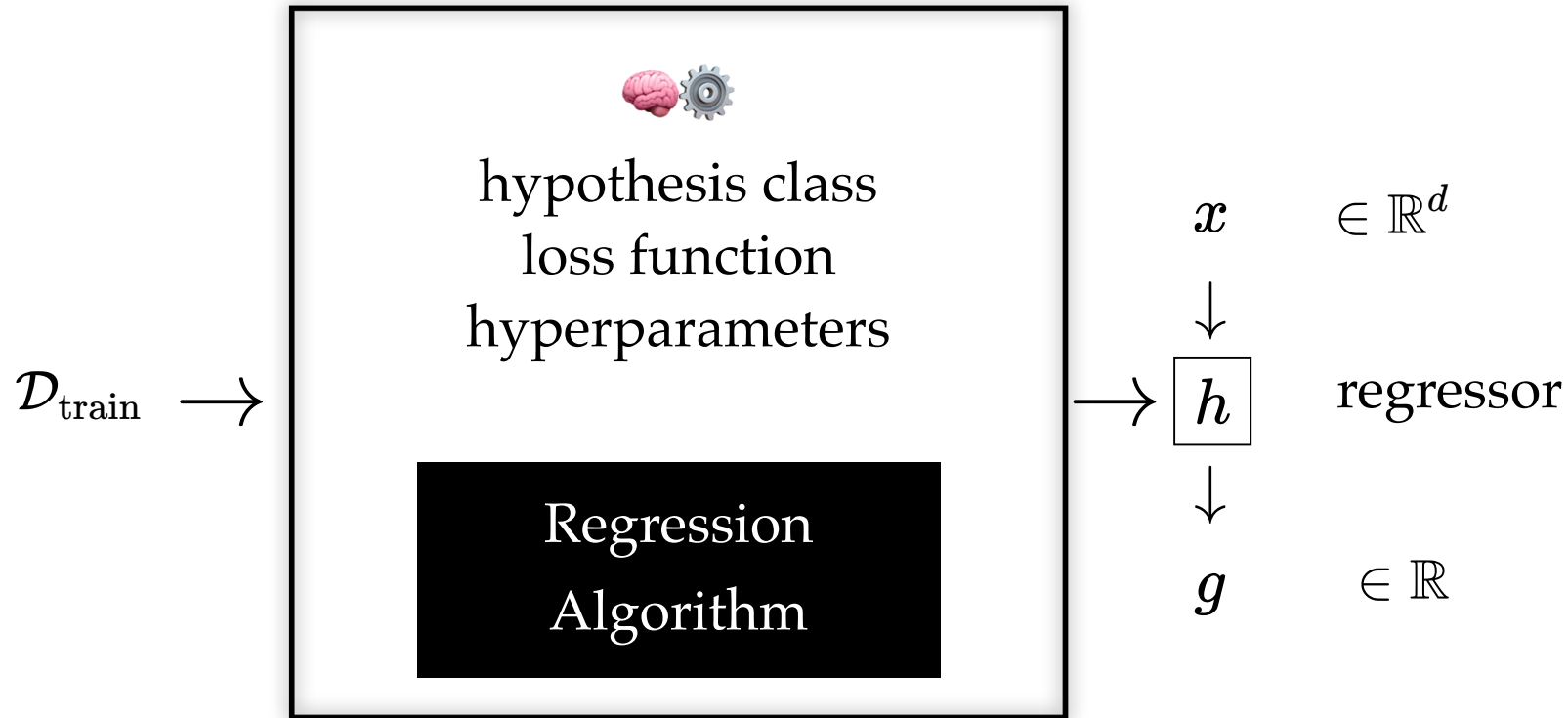
[Slides and Lecture Recording](#)

Recap:

"Learn" a model



Recap:



"Learn" a model $\rightarrow$

train, optimize, tune, adapt ...

adjusting / updating / finding θ

gradient based

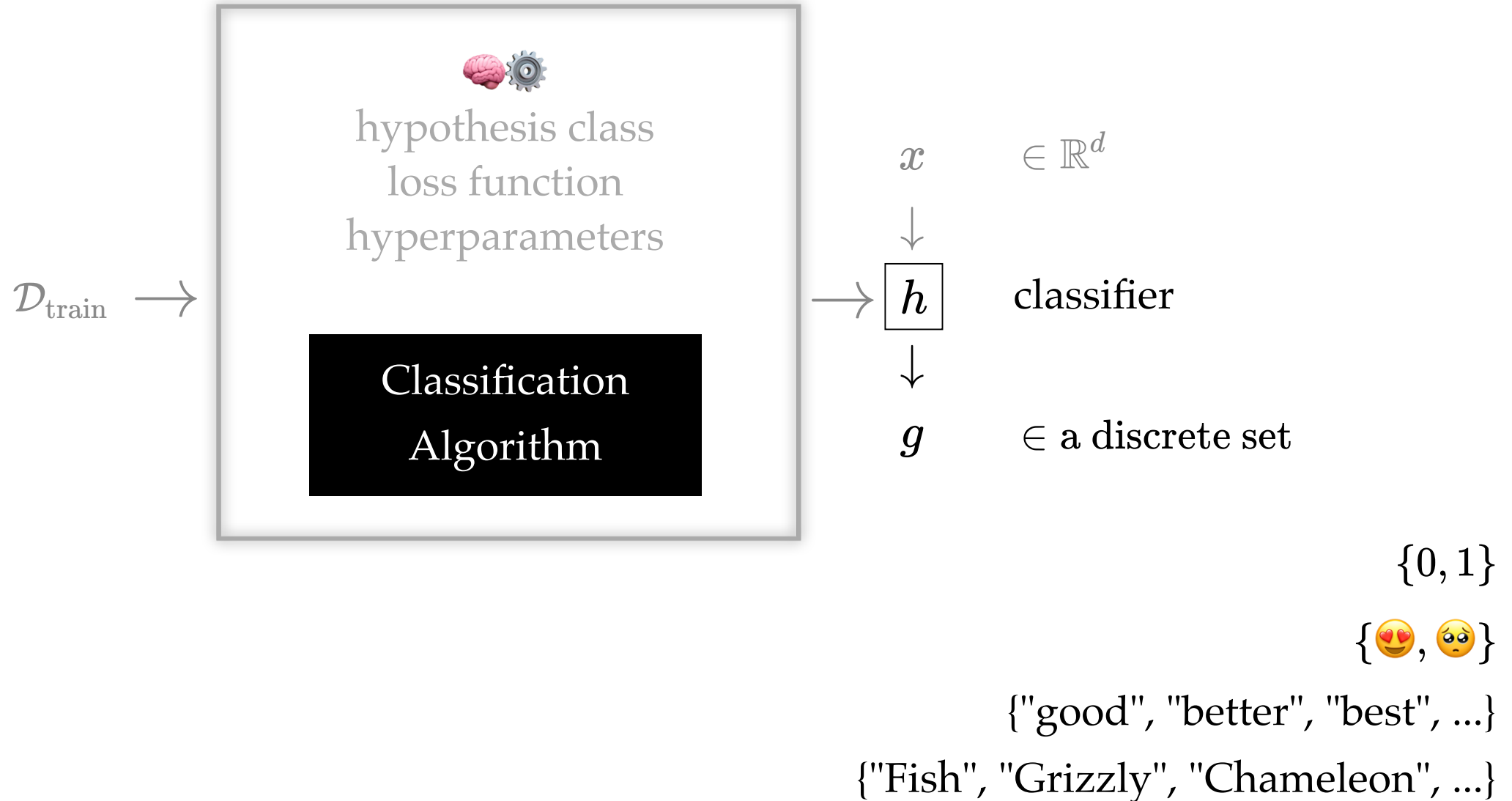
$\downarrow$ "Use" a model

predict, test, evaluate, infer ...

plug in the θ found

no gradients involved

Today:



Outline

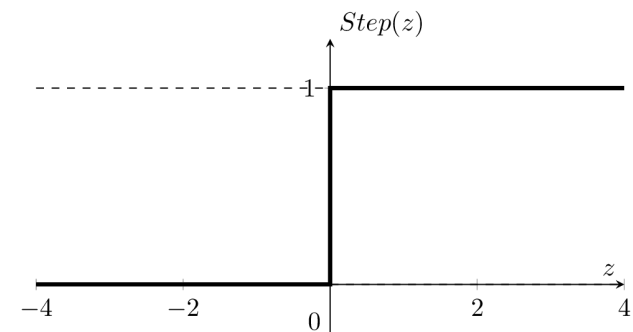
1. Linear (binary) classifiers

- to use: **separator, normal vector**
- to learn: very difficult!

2. Linear logistic (binary) classifiers

3. Linear multi-class classifiers

	linear regressor	linear binary classifier
features	$x \in \mathbb{R}^d$	
label	$y \in \mathbb{R}$	$y \in \{0, 1\}$
parameters	$\theta \in \mathbb{R}^d, \theta_0 \in \mathbb{R}$	
linear combination	$\theta^T x + \theta_0 = z$	
predict	$g = z$	$g = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{otherwise} \end{cases}$



today, we refer to $\theta^T x + \theta_0$ as z throughout.

<https://shenshen.mit.edu/demos/separator.html>

Outline

1. Linear (binary) classifiers

- to use: separator, normal vector
- to learn: very difficult!

2. Linear logistic (binary) classifiers

3. Linear multi-class classifiers

- To *learn* a model, need a loss function.

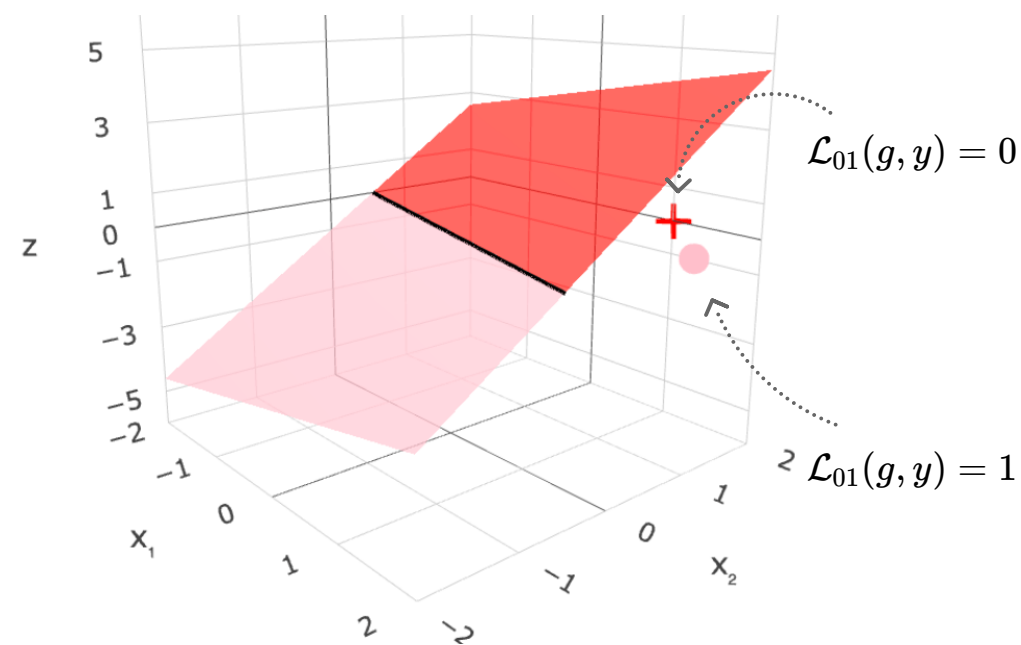
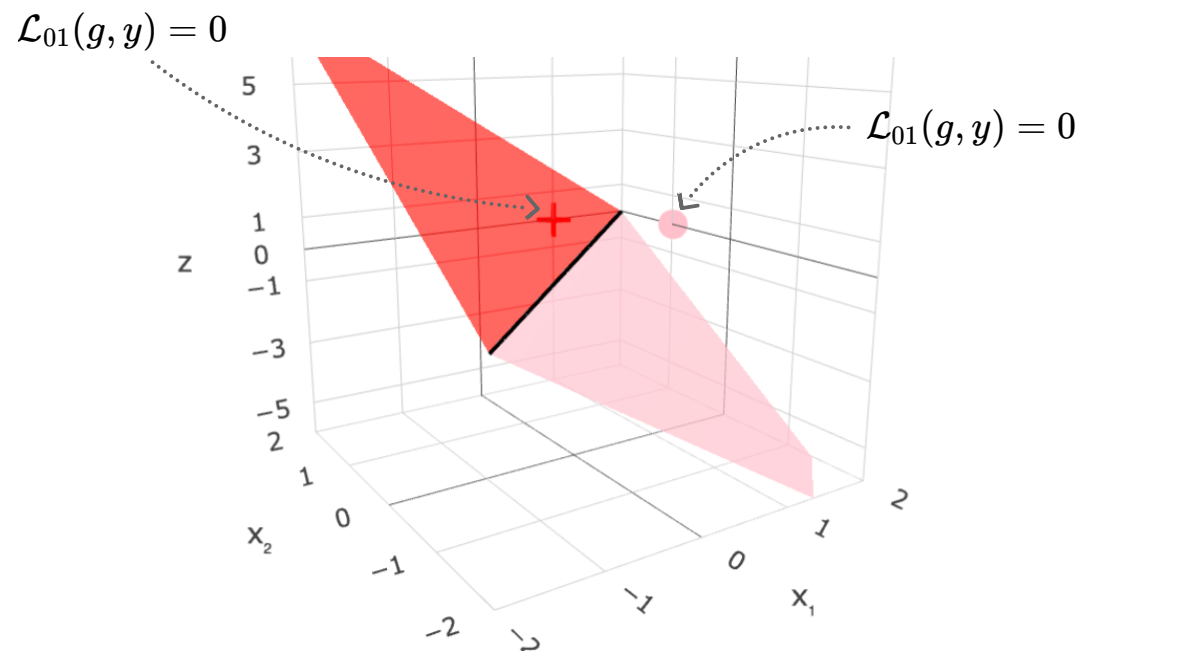
- One natural loss choice:

$$g = \text{step}(z) = \text{step}(\theta^\top x + \theta_0)$$

y

$$\mathcal{L}_{01}(g, y) = \begin{cases} 0 & \text{if guess} = \text{label} \\ 1 & \text{otherwise} \end{cases}$$

- Very intuitive, and easy to evaluate 🥰



<https://shenshen.mit.edu/demos/classification/ols-01-lr.html?models=ols,01>

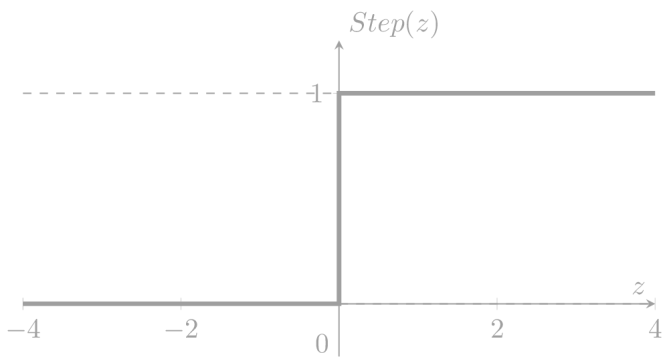
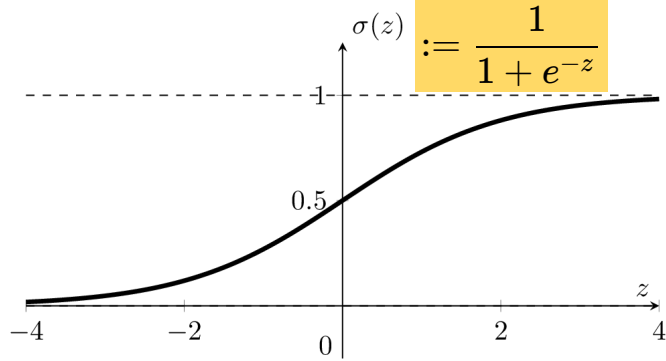
$J_{01}(\theta)$ very hard to optimize (NP-hard) 🙄

- "Flat" almost everywhere (zero gradient $\nabla_{\theta} J_{01}(\theta)$)
- "Jumps" elsewhere (no gradient)

	linear regressor	linear binary classifier	
features	$x \in \mathbb{R}^d$		
	$y \in \mathbb{R}$	$y \in \{0, 1\}$	
parameters	$\theta \in \mathbb{R}^d, \theta_0 \in \mathbb{R}$		
linear combo	$\theta^T x + \theta_0 = z$		
predict	$g = z$	$g = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{otherwise} \end{cases}$	g is "flat" and discrete in θ
loss $\mathcal{L}(g, y)$	$\mathcal{L}_{\text{squared}} = (g - y)^2$	$\mathcal{L}_{01} = \begin{cases} 0 & \text{if } g = y \\ 1 & \text{otherwise} \end{cases}$	$\mathcal{L}_{01}(g, y)$ is "flat" and discrete in g
optimize method	- closed-form formula - gradient descent	training error almost "flat" w.r.t θ , gradient gives very little info	

Outline

1. Linear (binary) classifiers
2. Linear logistic (binary) classifiers
 - to use: **sigmoid**
 - to learn: **negative log-likelihood loss**
3. Linear multi-class classifiers

	linear binary classifier	linear <i>logistic</i> binary classifier
features	$x \in \mathbb{R}^d$	
parameters	$\theta \in \mathbb{R}^d, \theta_0 \in \mathbb{R}$	
linear combo	$\theta^T x + \theta_0 = z$	
predict	 $\begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{otherwise} \end{cases}$	 $\begin{cases} 1 & \text{if } \sigma(z) > 0.5 \\ 0 & \text{otherwise} \end{cases}$ Sigmoid $\sigma(\cdot)$: <i>confidence</i> or <i>estimated likelihood</i> that x belongs to the positive class

https://shenshen.mit.edu/demos/classification/step_sigmoid.html

- θ, θ_0 can flip, squeeze, expand, or shift the $\sigma(x)$ graph *horizontally*
- $\sigma(\cdot)$ monotonic, very elegant gradient (see hw/lab)

linear logistic binary classifier

features: $x \in \mathbb{R}^d$

parameters: $\theta \in \mathbb{R}^d, \theta_0 \in \mathbb{R}$

the *logit* z :

$$z = \theta^\top x + \theta_0$$

apply *sigmoid*:

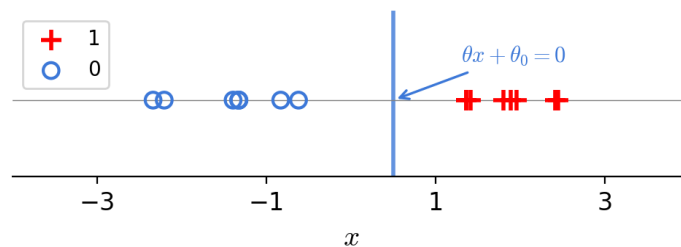
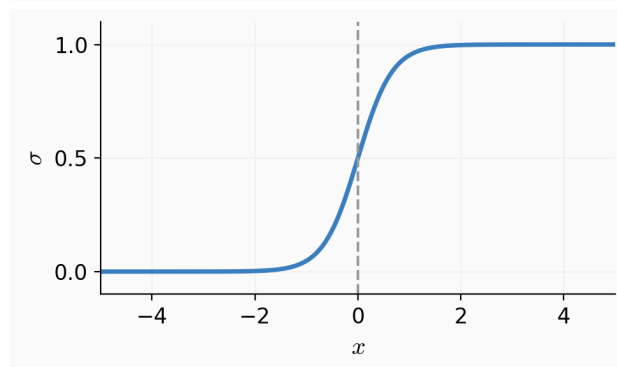
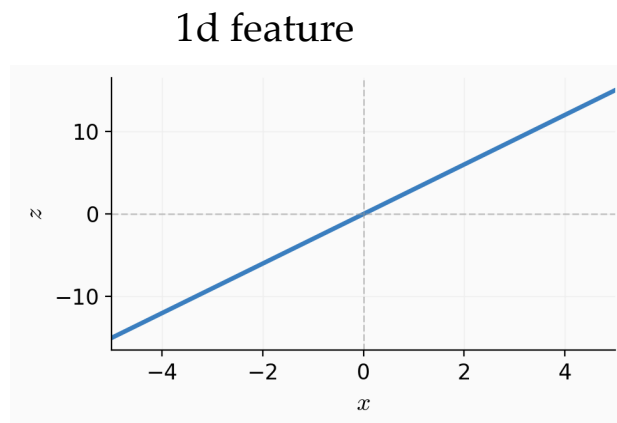
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Predict 1 if $\sigma(z) > 0.5$, else 0.

$$\sigma(z) = 0.5 \iff z = 0$$

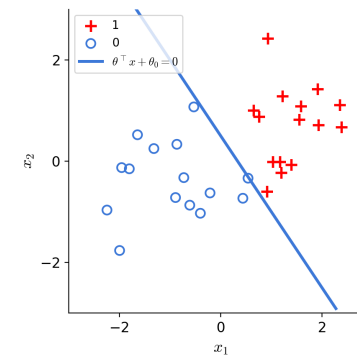
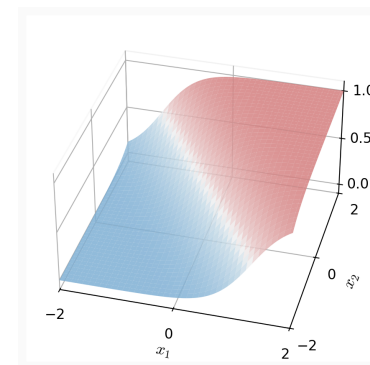
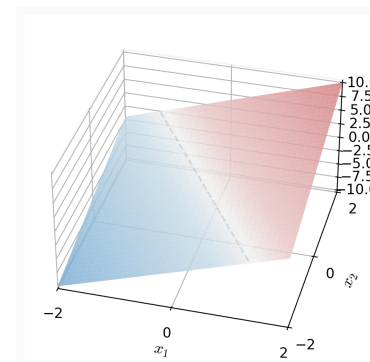
$$\iff \theta^\top x + \theta_0 = 0$$

T
e
x
t



T
e
x
t

2d features

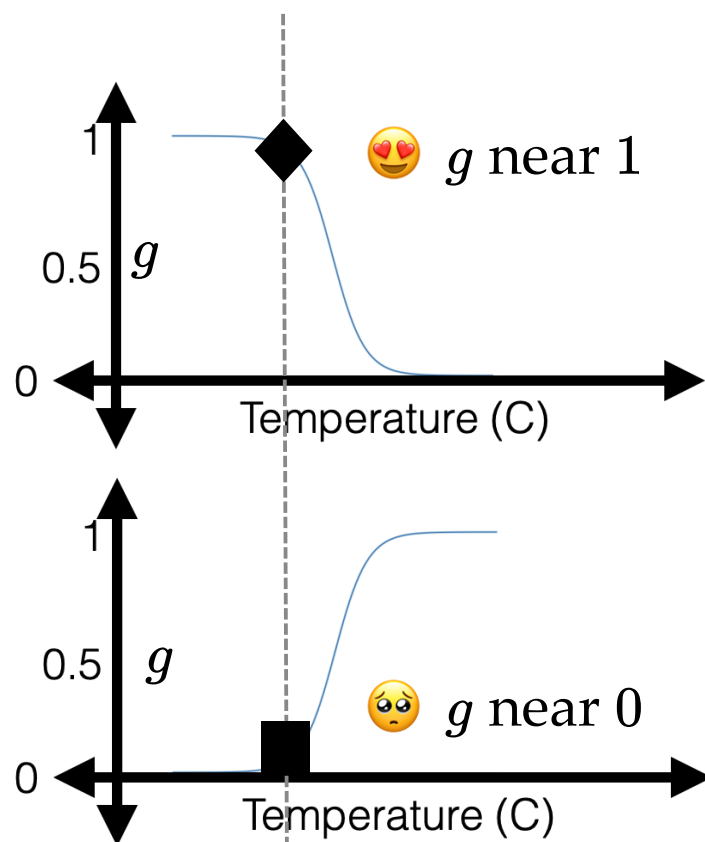
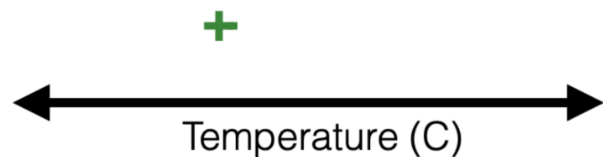


separator is *linear* in feature x !

Outline

1. Linear (binary) classifiers
2. Linear **logistic** (binary) classifiers
 - to use: **sigmoid**
 - to learn: **negative log-likelihood loss**
3. Linear multi-class classifiers

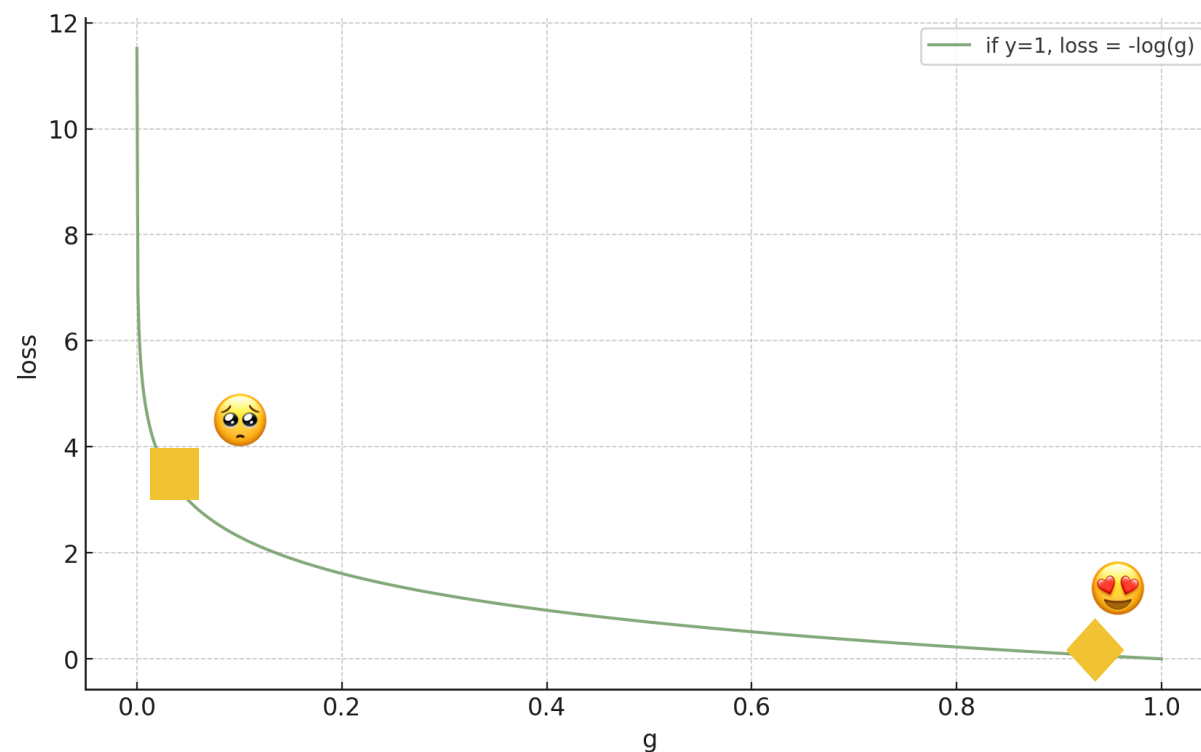
Example: a single training data from $y = 1$ class



want a smooth loss $\mathcal{L}(g, y)$ to reward g closer to y ?

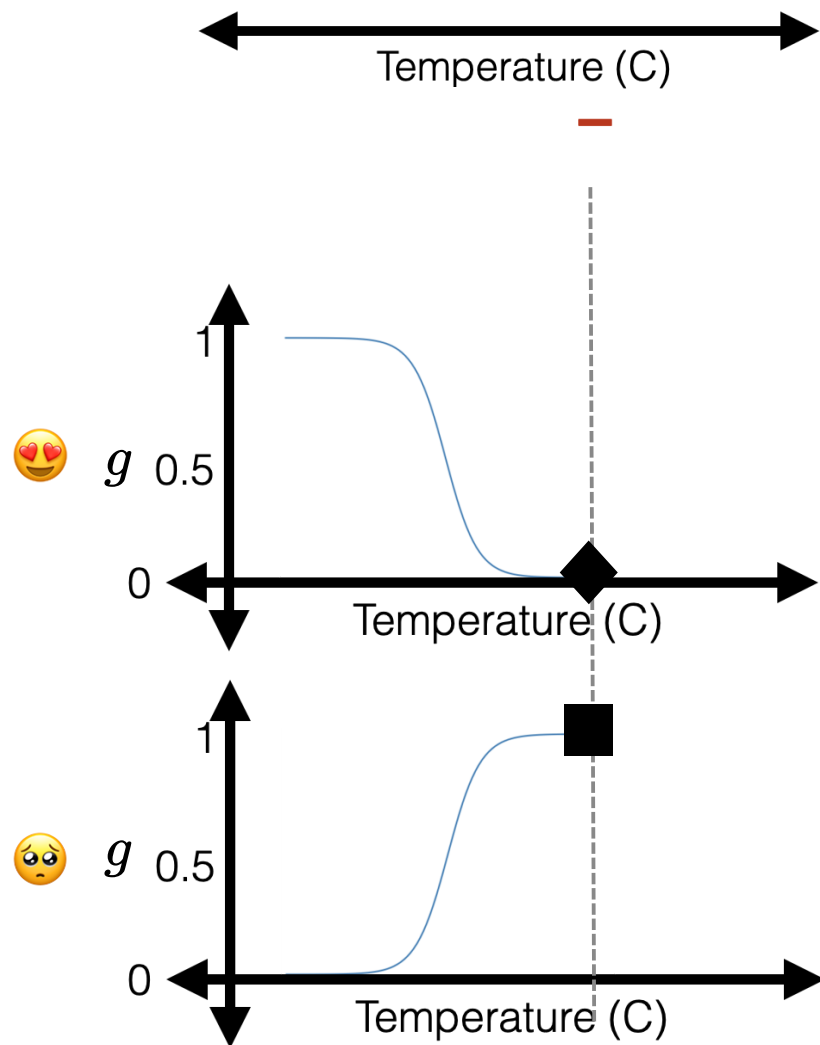
negative log likelihood

$$\mathcal{L}_{\text{nll}}(g, y) = -\log g$$

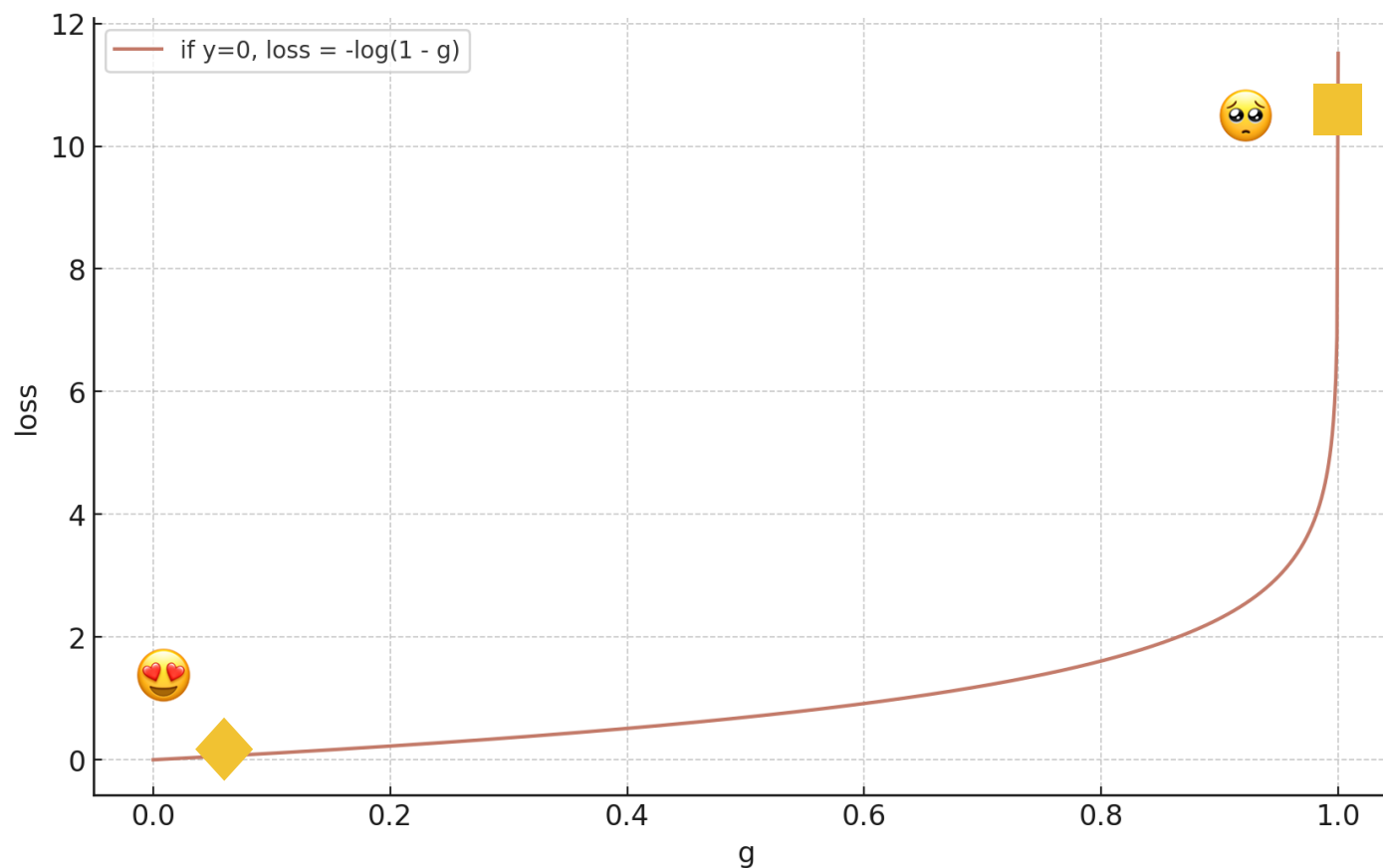


Example: a single training data from $y = 0$ class

$1 - g = 1 - \sigma(\cdot)$: the model's *predicted likelihood* that x belongs to the **negative** class.



$$\mathcal{L}_{\text{nll}}(g, 0) := -\log(1 - g)$$



<https://shenshen.mit.edu/demos/classification/nll.html>

Because the actual label $y \in \{0, 1\}$,

$$\mathcal{L}_{\text{nll}}(g, y) = \begin{cases} -\log(g) & \text{if } y = 1 \\ -\log(1 - g) & \text{if } y = 0 \end{cases} \Leftrightarrow -[y \log g + (1 - y) \log(1 - g)]$$

- When $y = 1$: $-[y \log g + (1 - y) \log(1 - g)] = -\log g$
- When $y = 0$: $-[y \log g + (1 - y) \log(1 - g)] = -\log(1 - g)$

Read as: $\sum (\text{true label for class } k) \cdot -\log(\text{predicted prob of class } k)$.

Since $y \in \{0, 1\}$, only the true class's term survives.

	linear binary classifier	linear <i>logistic</i> binary classifier
features	$x \in \mathbb{R}^d$	
label	$y \in \{0, 1\}$	
parameters	$\theta \in \mathbb{R}^d, \theta_0 \in \mathbb{R}$	
linear combo	$\theta^T x + \theta_0 = z$	
predict	$\begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{otherwise} \end{cases}$	$\begin{cases} 1 & \text{if } g = \sigma(z) > 0.5 \\ 0 & \text{otherwise} \end{cases}$
loss	$\begin{cases} 0 & \text{if } g = y \\ 1 & \text{otherwise} \end{cases}$	$\begin{cases} -\log(g) & \text{if } y = 1 \\ -\log(1 - g) & \text{if } y = 0 \end{cases}$ $\Leftrightarrow -[y \log g + (1 - y) \log(1 - g)]$
optimize via	NP-hard to learn	gradient descent

training data: $x = 1, y = 1$

$$\mathcal{L}_{\text{nll}}(g, 1) := -\log(g)$$

<https://shenshen.mit.edu/demos/nlloverfit.html>

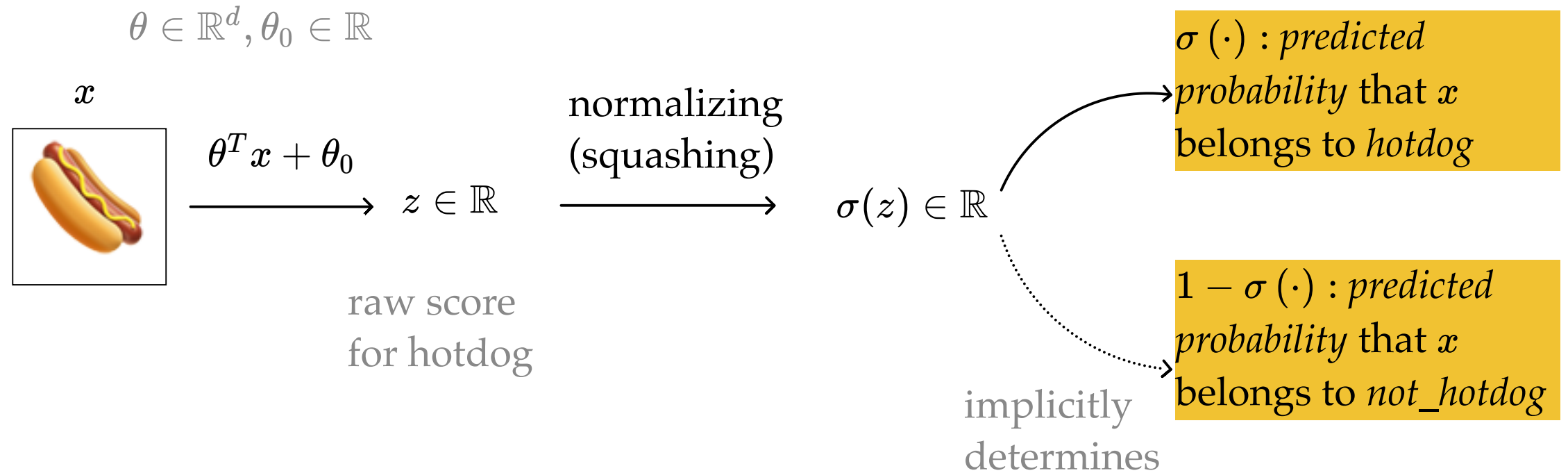
- If the data set is linearly separable, nll has no minimizer
- in theory, θ tends to have large magnitude $\Rightarrow$ overly confident
- common to add ridge penalty $\lambda \|\theta\|^2$

Outline

1. Linear (binary) classifiers
2. Linear logistic (binary) classifiers
3. Linear multi-class classifiers
 - to use: **softmax**
 - to learn: **one-hot encoding, cross-entropy loss**



to predict $\{hotdog, \text{or } not_hotdog\}$, a scalar logit z suffices

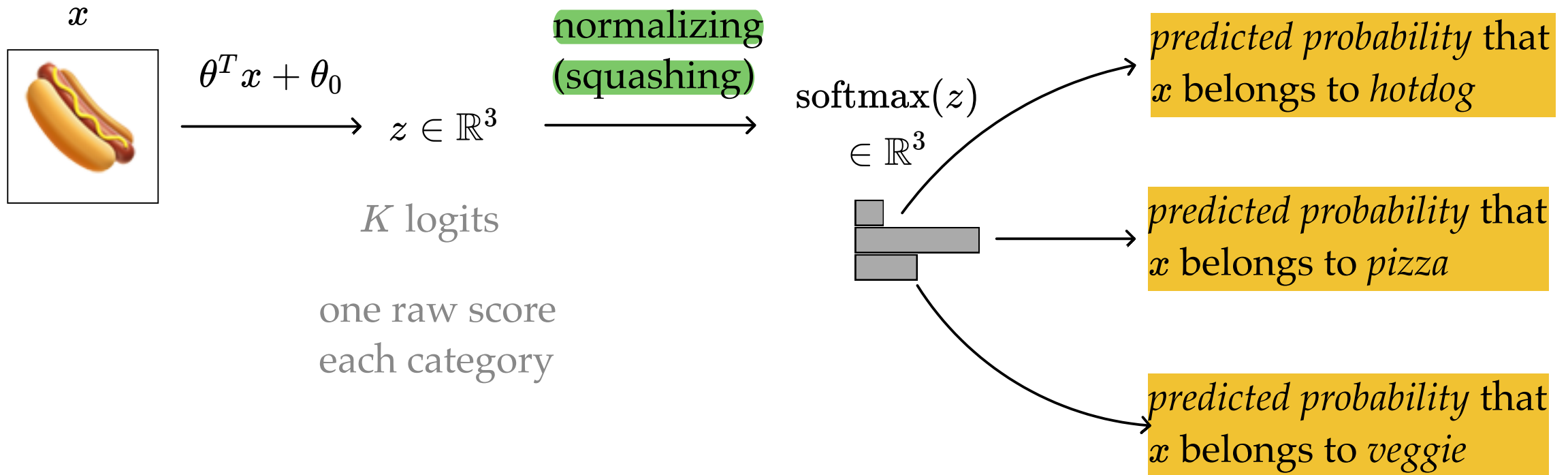


for $K > 2$ classes, a single scalar z no longer suffices — use K logit scores to keep track

for K classes, use K logit scores.

e.g. $K = 3$: $\{hot\text{-}dog, pizza, veggie\}$

$$\theta \in \mathbb{R}^{d \times K}, \theta_0 \in \mathbb{R}^K$$



softmax: $\mathbb{R}^K \rightarrow \mathbb{R}^K$

$$\text{softmax}(z) := \begin{bmatrix} \frac{\exp(z_1)}{\sum_{k=1}^K \exp(z_k)} \\ \vdots \\ \frac{\exp(z_K)}{\sum_{k=1}^K \exp(z_k)} \end{bmatrix}$$

outputs all $\in [0, 1]$,
sum to 1

e.g.

$$\text{softmax}\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} \frac{e^1}{e^1 + e^2 + e^3} \\ \frac{e^2}{e^1 + e^2 + e^3} \\ \frac{e^3}{e^1 + e^2 + e^3} \end{bmatrix} = \begin{bmatrix} 0.0900 \\ 0.2447 \\ 0.6653 \end{bmatrix}$$

max among the K logits

"soft" max'd in the output

sigmoid $\mathbb{R} \rightarrow \mathbb{R}$

$$\begin{aligned}\sigma(z) &:= \frac{1}{1 + \exp(-z)} \\ &= \frac{\exp(z)}{\exp(z) + \exp(0)}\end{aligned}$$

implicit logit for the negative class

predict positive if $\sigma(z) > 0.5 = \sigma(0)$

softmax: $\mathbb{R}^K \rightarrow \mathbb{R}^K$

$$\text{softmax}(z) := \begin{bmatrix} \frac{\exp(z_1)}{\sum_{k=1}^K \exp(z_k)} \\ \vdots \\ \frac{\exp(z_K)}{\sum_{k=1}^K \exp(z_k)} \end{bmatrix}$$

predict the category with the highest softmax score

unifying rule: predict the class with the largest logit

	linear logistic <i>binary</i> classifier	one-out-of- K classifier
features	$x \in \mathbb{R}^d$	
parameters	$\theta \in \mathbb{R}^d, \theta_0 \in \mathbb{R}$	$\theta \in \mathbb{R}^{d \times K}, \theta_0 \in \mathbb{R}^K$
linear combo	$\theta^T x + \theta_0 = z \in \mathbb{R}$	$\theta^T x + \theta_0 = z \in \mathbb{R}^K$
predict	$\sigma(z) = \frac{\exp(z)}{\exp(0) + \exp(z)}$ predict positive if $\sigma(z) > \sigma(0)$	$\text{softmax}(z) = \begin{bmatrix} \frac{\exp(z_1)}{\sum_{k=1}^K \exp(z_k)} \\ \vdots \\ \frac{\exp(z_K)}{\sum_{k=1}^K \exp(z_k)} \end{bmatrix}$ predict the class with the highest softmax score

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1. Linear (binary) classifiers
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3. Linear multi-class classifiers
 - to use: **softmax**
 - to learn: **one-hot encoding**, **cross-entropy loss**

One-hot encoding:

- Generalizes from $\{0, 1\}$ binary labels
- Encode the K classes as an $\mathbb{R}^K$ vector, with a single 1 (*hot*) and 0s elsewhere

Training data

x	y
( , "hot-dog")	
( , "pizza")	
( , "veggie")	
( , "veggie")	
$\vdots$	

$K = 3$



Training data

x	y
( , $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$)	
( , $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$)	
( , $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$)	
( , $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$)	
$\vdots$	

in general, for K classes:

class:

1	2	$\dots$	K
		$\dots$	
		$\dots$	
		$\dots$	
		$\dots$	
		$\dots$	

Negative log-likelihood K – classes loss (aka, cross-entropy)

g : softmax output

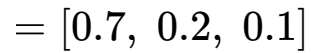
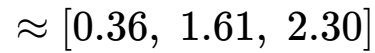
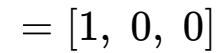
g_k : probability or confidence of belonging in class k

$$\mathcal{L}_{\text{nllm}}(g, y) = - \sum_{k=1}^K y_k \cdot \log(g_k)$$

y : one-hot encoding label

y_k : k th entry in y , either 0 or 1

- Generalizes negative log likelihood loss $\mathcal{L}_{\text{nll}}(g, y) = -[y \log g + (1 - y) \log (1 - g)]$
- Despite the K – term sum, only the term corresponding to its true class label contributes, since all other $y_k = 0$

A hot dog with mustard.
$$-\log(g)$$
 y (true label)

Loss ≈ 0.36

33

Summary

- Classification predicts a label from a discrete set; a linear binary classifier separates the feature space with a hyperplane defined by θ, θ_0 .
- The 0-1 loss is natural for classification but NP-hard to optimize.
- The sigmoid $\sigma(z)$ gives a smooth, probabilistic step function; paired with the NLL loss, we can train via (S)GD.
- Regularization remains important for logistic classifiers.
- Multi-class classification generalizes via one-hot encoding and softmax.